

# Fenyong Li

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## PROFILE

A highly adaptable professional who consistently drives impact in any new environment. Ranked 1st in both my undergraduate and graduate cohorts, I secured top academic honors despite transitioning to a completely new major. Driven by a deep passion for real-world industry applications, I chose to decline a fully-funded U.S. Ph.D. offer and a Master's admission to Nanyang Technological University. **My core strengths lie in my quick learning ability, adaptability, and excellent teamwork, along with a strong willingness to learn and grow with others.**

*"I firmly believe that the AI revolution has fundamentally disrupted traditional education, ensuring that mastering any field of knowledge is no longer a barrier."*

## EDUCATION

- Guilin University of Electronic Technology** Guangxi, China  
*Bachelor of Science in Computer Science and Technology* Sep. 2020 - Jun. 2024
  - CGPA: 89.4/100 (**Ranked 1<sup>st</sup> in Major**)
  - Key Courses:** Course Design of Computer Composition Principle (100/100), Advanced Mathematics (98/100), Data Structure & Algorithm (97/100), Operating System(96/100)...
- University of Malaya** Kuala Lumpur, Malaysia  
*Master of Computer Science* Mar. 2025 - Dec. 2026 (*Expected*)
  - CGPA: 3.90/4.00 (**Ranked 1<sup>st</sup> in Major**)
  - Key Courses:** Cloud Computing (A+), Augmented Reality (A+), Advanced Algorithms (A), Advanced Machine Learning (A)...

## PUBLICATIONS

- Li, Fenyong, 2023.:** "Y-Net: Convolutional Networks for Multi-Domain Image Segmentation." *2023 5th International Conference on Electronic Engineering and Informatics (EEI)*. IEEE, 2023.
  - Abstract:** We developed Y-Net, a refined image segmentation algorithm, to tackle the issue of inadequate local detail and global contrast extraction in existing SOD networks. By integrating the advanced U2NetP and RAS Net within deep learning, and leveraging a custom module and residual mechanism, Y-Net excels in segmenting diverse image types. It outperforms the foundational networks across five major public datasets.
- Li, Fenyong. 2023.:** "A Machine Learning Device for Rapidly Improving Image Recognition Accuracy." China Patent Application 202311098188.3. Patent Pending.
- Li, Fenyong. 2024.:** "An Integrated Computer." China Patent Application 202410382065.0. Patent Pending.

## INTERNSHIP

- Malaysia-Zhejiang International Investment Association** Kuala Lumpur, Malaysia  
*Technical Consultant (AI & Automation Solutions)* Mar. 2025 - Dec. 2025
  - LLM-Driven Agentic Workflows & RAG Architecture:** Architected a Retrieval-Augmented Generation (RAG) pipeline leveraging **LangChain** to process high volumes of unstructured meeting minutes. Developed autonomous agent workflows utilizing precise prompt engineering and LCEL (LangChain Expression Language) for custom tool-calling, achieving a **95%+ accuracy** in structured data extraction.
  - Autonomous Data Ingestion Agents:** Developed and deployed Python-based autonomous scraping agents for continuous multi-source data collection. Integrated these agents with LLMs to seamlessly process **10,000+ data points weekly**, providing real-time, data-driven analytical support and fully automating the generation of executive reports.
- Glorybene Smart Systems Pte. Ltd.** Singapore  
*System Engineer* Mar. 2023 - Dec. 2023
  - NALA+ System Development:** Contributed significantly to the end-to-end development of the NALA+ IoT gateway, covering requirement analysis, architectural design, core control logic, and production delivery. Managed cross-platform hardware driver adaptation and low-level protocol stack implementation, ensuring system stability in complex overseas industrial environments.
  - Communication Architecture & Optimization:** Redesigned and implemented an asynchronous communication mechanism for distributed nodes to resolve network congestion and packet loss during high-concurrency sensor data ingestion. Conducted extensive stress testing and performance tuning, reducing system response latency by nearly 30%.
  - JTC Project Implementation:** Spearheaded energy simulation and energy-saving strategy iterations for Singapore Jurong Town Corporation (JTC) office buildings. Optimized lighting control algorithms to reduce Lighting Power Density (LPD) to 5W/m<sup>2</sup>, helping the project achieve the highest energy efficiency standards set by the Building and Construction Authority (BCA).

HONORS & AWARDS

- Scholarships & Academic Honors
  - Provincial Government Scholarship | Provincial Department of Education Top 0.24%
  - Lingyi Tech Corporate Scholarship | Lingyi iTech Co., Ltd. Top 0.73%
  - Outstanding Undergraduate Thesis Award | Guilin University of Electronic Technology Top 1.10%
  - Provincial Outstanding Graduate | Provincial Department of Education Top 1.34%
  - First-Class Academic Scholarship & Model Student | Guilin University of Electronic Technology Top 1.34%
  - Other Honors: 10+ awards for academic, leadership, and volunteer excellence.
- Technical Competitions
  - Top 10 Finalist Global AI Hackathon Singapore | Tencent Cloud 2026
  - Provincial 3rd Prize ACM-ICPC Programming Contest (GXCP) | Provincial Department of Education 2022
  - National 3rd Prize Collegiate Network Tech Challenge | MOE Computer Education Association 2024
  - National Excellence Software & IT Professionals Competition | Ministry of Industry & Info Tech 2022
  - Provincial 2nd Prize Int'l Innovation & Entrepreneur Contest | Provincial Department of Education 2024
  - Provincial 2nd Prize "Challenge Cup" Academic & Tech Contest | Provincial Department of Education 2023
  - Provincial 2nd Prize National Mathematical Modeling (CUMCM) | China Society for Ind. & Applied Math 2022

PROJECTS

- Multimodal AR AI English Tutor | Tech Stack: Python, LangChain, RAG, LLM, VectorDB May 2026  
Architected an autonomous multimodal AI tutoring agent, leveraging Augmented Reality to establish real-time visual perception and spatial awareness of the user's physical environment. Engineered a cognitive decision-making core using GPT, incorporating Chain-of-Thought (CoT) prompting to dynamically analyze user interactions and autonomously formulate personalized pedagogical strategies. Designed a multi-tier memory architecture (short-term dialogue buffering and long-term user profile retrieval) to maintain context continuity and adapt to the user's evolving learning curve. Optimized the Python-Unity integration pipeline to instantly translate the agent's logical decisions into synchronized, multimodal executions, including fluid voice output and corresponding 3D avatar animations.
- Generative VLM Evaluation & Reasoning | Tech Stack: PyTorch, HuggingFace, LLM, VLM Jan 2026  
Spearheaded a comparative evaluation between a discriminative CNN-LSTM baseline and an advanced generative BLIP framework to develop robust perception layers for clinical AI assistants. Leveraged Kaggle's high-performance P100 GPUs and the Hugging Face Transformers ecosystem to orchestrate complex multimodal training pipelines. Engineered a visual-linguistic reasoning core using Vision Transformers (ViT) and Cross-Attention mechanisms. Successfully overcame the traditional "classification trap" via auto-regressive token generation, significantly improving the agent's semantic reasoning (F1 Score) and zero-shot Out-of-Vocabulary (OOV) capabilities.
- Automated Farmland Management System | Tech Stack: Flask, Streamlit, MicroPython, IoT Jun 2024  
Architected a full-stack "Edge-Cloud" system for precision agriculture. Integrated CV-based facial authentication and millisecond-level fire warnings. Developed a sensor-driven intelligent "Fertigation" system. Synergized UGV autonomous inspection with fixed stations for full-dimensional monitoring, achieving closed-loop management. Awarded "Outstanding Graduation Project".
- Ultra-Lightweight Sign Language Translator | Tech Stack: YoloSeg, CV Algorithms, Android Native Apr 2023  
Developed a portable translation tool utilizing a miniature camera and YoloSeg for real-time gesture recognition. Optimized for mobile deployment via operator acceleration and cross-platform instruction tuning to ensure high-precision semantic output. Project fully funded by research grants; published 1 IEEE conference paper and obtained a National Software Copyright.
- Behavior Tree AI Agent Game Framework | Tech Stack: C#, Unity 3D, Client-Server State Sync Mar 2021  
Designed a high-extensibility RPG core framework using Unity 3D. Developed a complex NPC Behavior Tree system and a dynamic quest-driven system to support flexible AI interactions and branching logic. Independently implemented a reliable Client-Server State Synchronization (State Sync) mechanism to ensure multi-terminal consistency. Optimized system robustness through decoupled inventory and event-driven architectures. Recruited as the first core member of the university's top-tier tech society in recent years due to exceptional technical execution.

LEADERSHIP

- University Science and Technology Association GUET  
Head of Software Department Sep 2021 - Jun 2024
  - Competition Orchestration & Management: Lead the planning of 6 university-level technical competitions and served as the lead evaluator. Spearheaded multi-channel outreach strategies attracting 1,600+ students and conducted 200+ technical interviews for high-caliber recruitment.
  - Technical Training & Team Leadership: Architected a milestone-based R&D training system from scratch and introduced Agile methodologies. Significantly enhanced the department's technical foundation and project delivery rate, leading the association to win the "Top 10 Club" honor.
- Grade Youth League General Branch GUET  
Vice Secretary Sep 2022 - Sep 2023
  - Organizational Operations & Resource Scheduling: Led a 30-member core team as the primary person in charge, managing a framework for nearly 1,000 students. Orchestrated college-level events, multi-dimensional scholarship evaluations, and critical safety risk control affairs.
  - Digital Workflow Optimization: Introduced cloud-based digital collaborative SOPs for task and progress tracking, significantly cutting communication costs and boosting management efficiency for nearly 1,000 students.